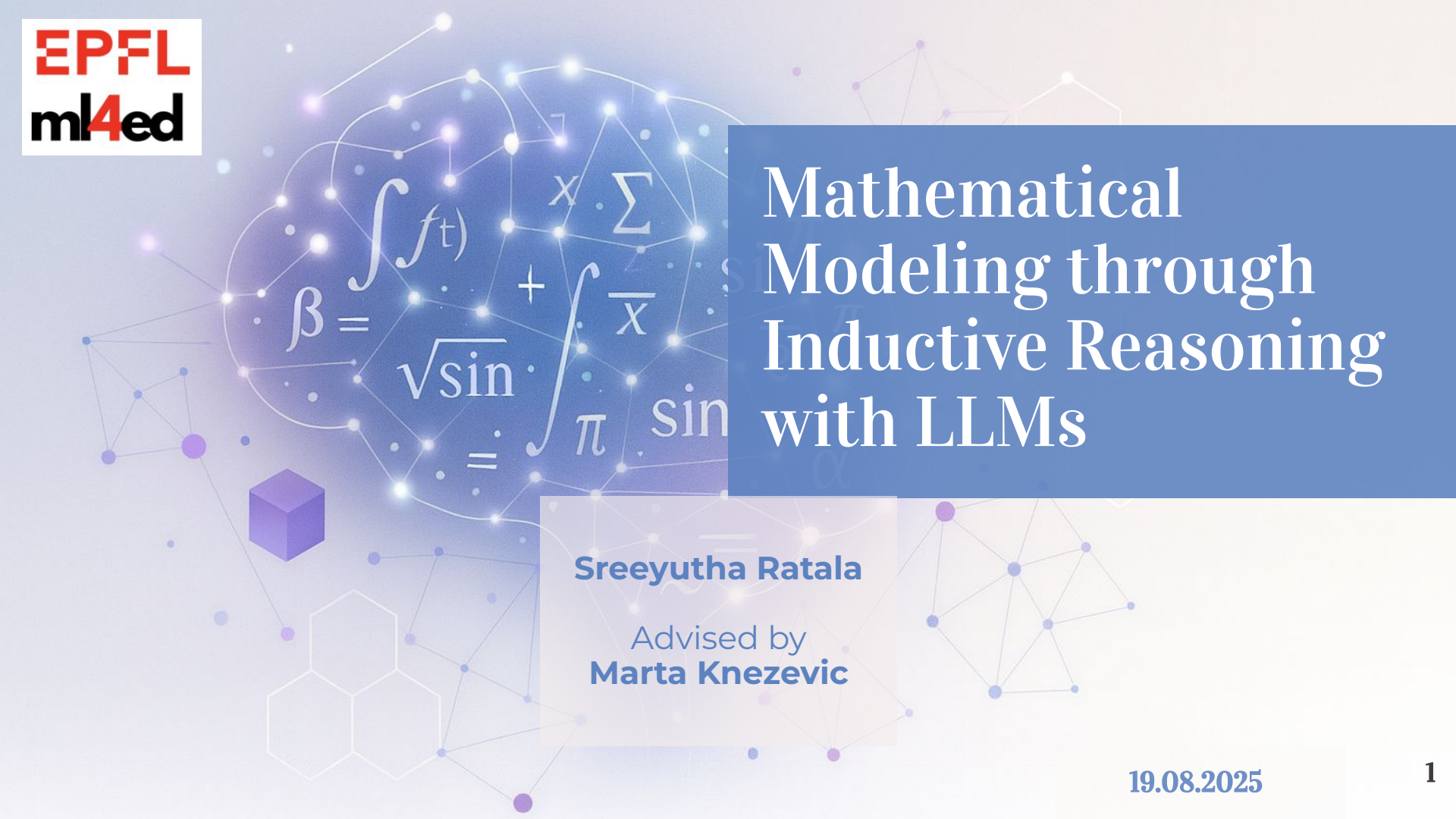


A complex background network diagram with nodes and edges, overlaid with mathematical symbols like β , $\int f_t$, $\sum$, $\sqrt{\sin}$, π , and $\sin$.

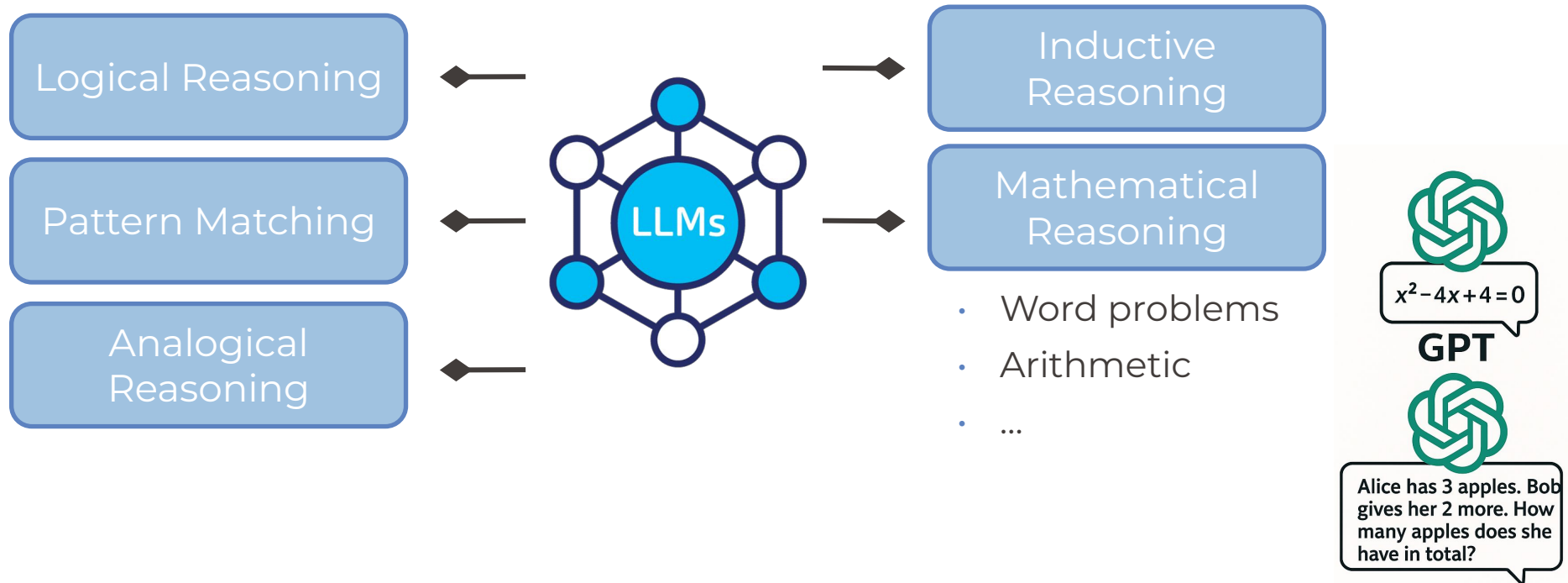
Mathematical Modeling through Inductive Reasoning with LLMs

Sreeyutha Ratala

Advised by
Marta Knezevic

Motivation

Why Study Math with LLMs?



How do LLMs process
numerical data?
How do LLMs do Math?

Memorization?

Reasoning?

Guessing?

Smth else?

Inductive
Reasoning

Mathematical
Reasoning

- Word problems
- Arithmetic
- ...



$$x^2 - 4x + 4 = 0$$

GPT



Alice has 3 apples. Bob gives her 2 more. How many apples does she have in total?

Our Task

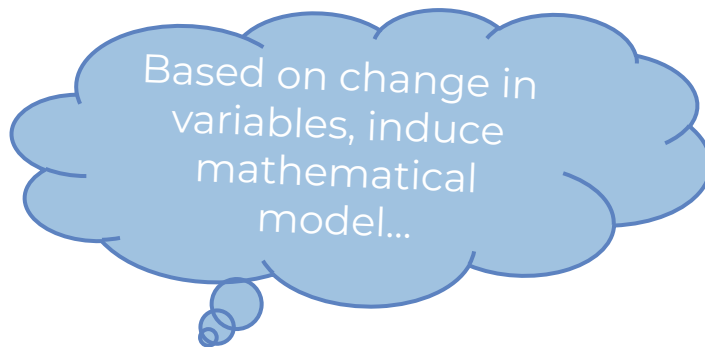
Numerical Modeling Through Inductive Reasoning

Novel Task

✗ Memorization ✓ Reasoning
(Inductive + Mathematical)

Given data points, what is the equation for x in terms of A , B , C ?

A	B	C	x
2	4	5	11
7	4	5	16
7	5	5	17



Ground
Truth:
 $x = A + B + C$

Data Points → Identify Patterns → Generate Expression → Evaluate Accuracy

Numerical Mathematical Puzzles/Games

- Countdown
- 24 Game

Mathematical Modeling for Word Problems

- Optimization Problems
- Differential Problems

Symbolic Regression

Given numeric data, find symbolic equations that model it.

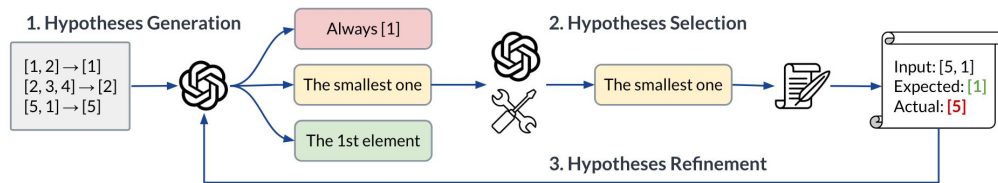
Inductive Reasoning through Hypothesizing

(Iterative) generation & evaluation of hypotheses to infer rules/patterns from examples.

Inductive Reasoning through Hypothesizing

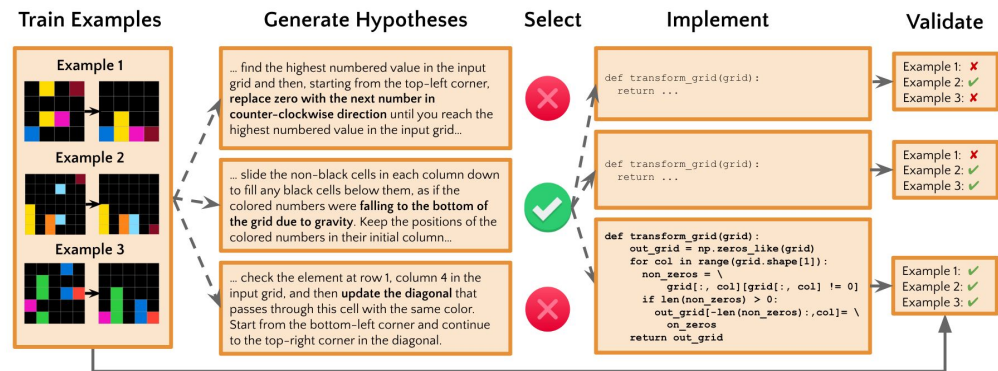
Iterative Hypothesis Refinement [1]

1. Hypothesis Generation
2. Hypothesis Selection
 - a. Task-specific symbolic interpreter
3. Hypothesis Refinement
4. Repeat



Hypothesis Search [2]

1. Hypothesis Generation
2. Hypothesis Selection
3. Hypothesis Implementation
4. Validation on Train Examples



*Pipelines shown, each starts with data samples

[1] [Phenomenal Yet Puzzling: Testing Inductive Reasoning Capabilities of Language Models with Hypothesis Refinement](#)

[2] [Hypothesis Search: Inductive Reasoning with Language Models](#)

Inductive Reasoning through Hypothesizing

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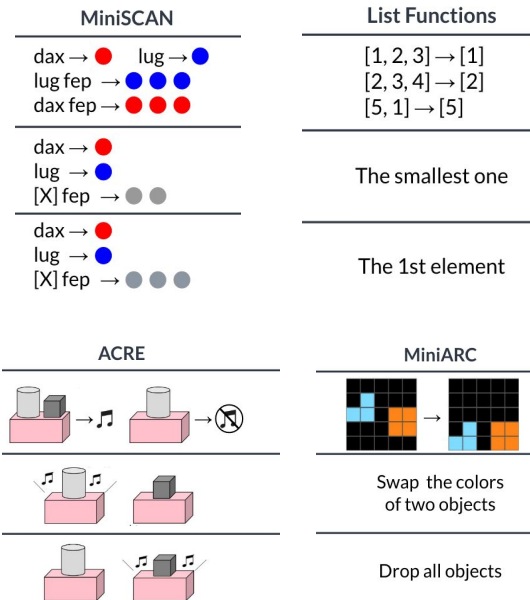
Hypothesis Search [2]

1. Hypothesis Generation
2. Hypothesis Selection
3. Hypothesis Implementation
4. Validation on Train Examples

*Pipelines shown, each starts with data samples

Inductive Reasoning Tasks

Symbolic, Visual, Logical (not numerical, like ours)



[1] [Phenomenal Yet Puzzling: Testing Inductive Reasoning Capabilities of Language Models with Hypothesis Refinement](#)

[2] [Hypothesis Search: Inductive Reasoning with Language Models](#)

Related Work

Numerical Mathematical Puzzles/Games

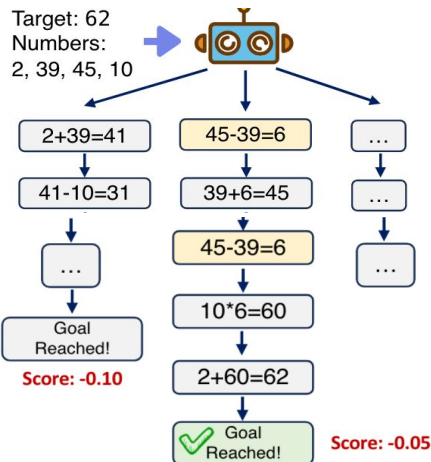


Combinatorial
Search

Inductive
Reasoning
(like ours)

Countdown [3, 4]

Use *any combination* of input numbers and basic operations (+, −, ×, ÷) to reach a target.



Game of 24 [5]

Use all input numbers *exactly once* with operations to make 24.

Input	4 numbers (4 9 10 13)
Output	An equation to reach 24 $(13-9)*(10-4)=24$

only one valid
solution

binary score

[3] [Stream of Search \(SoS\): Learning to Search in Language](#)

[4] [Step Back to Leap Forward: Self-Backtracking for Boosting Reasoning of Language Models](#)

[5] [Tree of Thoughts \(ToT\): Deliberate Problem Solving with Large Language Models](#)

Related Work

Mathematical Modeling for Word Problems

- Natural language $\rightarrow$ mathematical model
- Optimization problems & Differential equations
- Ex from [6]: Multiple valid equations for same physical scenario

Q: A car with an initial velocity of 0 m/s , and acceleration 1 m/s^2 , what is its displacement after 5 s ?	
M_1 : $y'(0) = 0$ $y''(t) = 1$ $A = y(5)$	M_2 : $v(0) = 0$ $v'(t) = 1$ $A = \int_0^5 v(s) ds$

Figure 1: Q is the natural language problem, while M_1, M_2 are different mathematical models.

Related Work

Mathematical Modeling for Word Problems

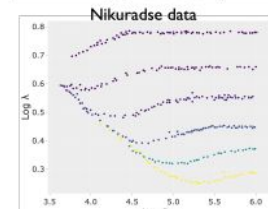
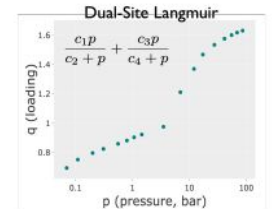
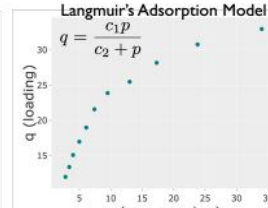
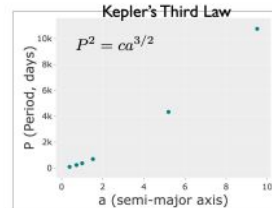
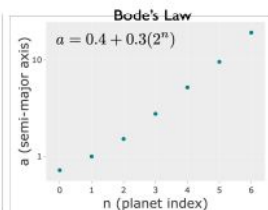
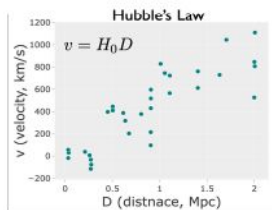
- Natural language $\rightarrow$ mathematical model
- Optimization
- ...

theirs:
language $\rightarrow$ math model
ours:
data points $\rightarrow$ math model

... is a word problem, while
... are different mathematical models.

Related Work

Symbolic Regression [7,8,9,10]



Dataset	Context
Langmuir	The data is about nitrogen adsorbing onto mica where the independent variable (x1) is pressure, and the dependent variable (y) is loading.
Dual Site Langmuir	The data is about isobutane adsorbing onto silicalite at constant temperature, where the independent variable (x1) is pressure, and the dependent variable (y) is loading.
Hubble	The data is about physical cosmology where the independent variable (x1) is proper distance to a galaxy, and the dependent variable (y) is its speed of separation.

Prompt 1

```

''' p,q
34,33
23.8,30.8
17.3,28.2
13,25.5
'''

```

Find an expression that best fits the data in CSV format separated by triple backticks.

GPT-4 output

The given data seems to follow a logarithmic pattern. From analyzing the data, the relationship between `p` and `q` can be approximated by the following expression:

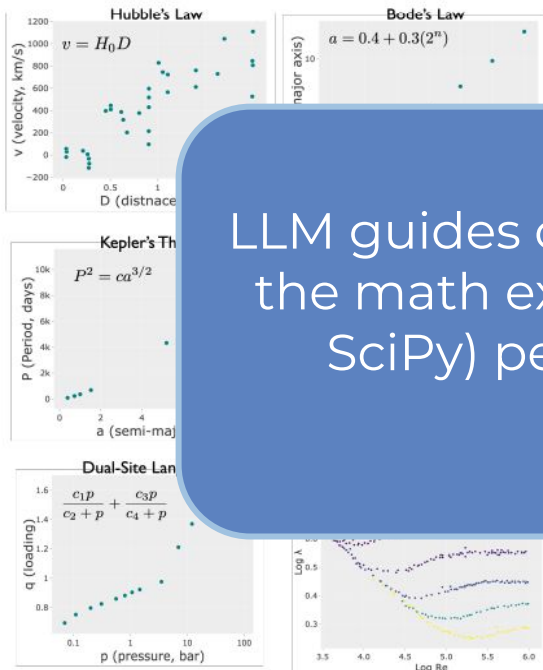
$$q = -9.692 * \ln(p) + 47.41$$

This equation was derived using logarithmic regression, and it should be a good fit for the given dataset. Please note, however, that there may be slight discrepancies due to the nature of regression analysis.

- [7] [In Context Learning and Reasoning for Symbolic Regression with Large Language Models](#)
- [8] [Symbolic Regression with a Learned Concept Library](#)
- [9] [In-Context Symbolic Regression: Leveraging Language Models for Function Discovery](#)
- [10] [Automated Statistical Model Discovery with Language Models](#)

Related Work

Symbolic Regression [7,8,9,10]



LLM guides optimization of the symbolic structure of the math expressions, while external modules (e.g., SciPy) perform numerical optimization of the constants.

Dataset	Context
Langmuir	The data is about nitrogen adsorbing onto mica where the independent variable (x1) is pressure, and the dependent variable (y1) is the amount of nitrogen adsorbed.
Kepler's Third Law	where a is the semi-major axis and P is the period. The data is about the relationship between the semi-major axis and the period of a planet.
Dual-Site Langmuir Adsorption	where q is the loading and p is the pressure. The data is about the relationship between the pressure and the loading of a gas on a solid surface.

Find an expression that best fits the data in CSV format separated by triple backticks.

for the given dataset. Please note, however, that there may be slight discrepancies due to the nature of regression analysis.

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Related Work

Symbolic Regression [7,8,9,10]

theirs :

- no clear inductive signal
- almost impossible to deduce model from only examples, without priors
- external tool usage

ours:

- inductive signal obvious
- no need for priors or external tools to deduce model

- [7] [In Context Learning and Reasoning for Symbolic Regression with Large Language Models](#)
[8] [Symbolic Regression with a Learned Concept Library](#)
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Research Questions

Goal: Investigate the **inductive reasoning** abilities of large language models (LLMs) in low-level numerical **mathematical modeling**.



RQ1: Baseline Performance

How well can LLMs reason inductively from numerical data **on their own**?



RQ3: Discoveries

Where do models **struggle** and what **reasoning bottlenecks** emerge?



RQ2: In-Context Learning

How can **in-context learning (ICL)** and **structured reasoning strategies** be used to improve LLMs' inductive reasoning in numerical mathematical modeling?



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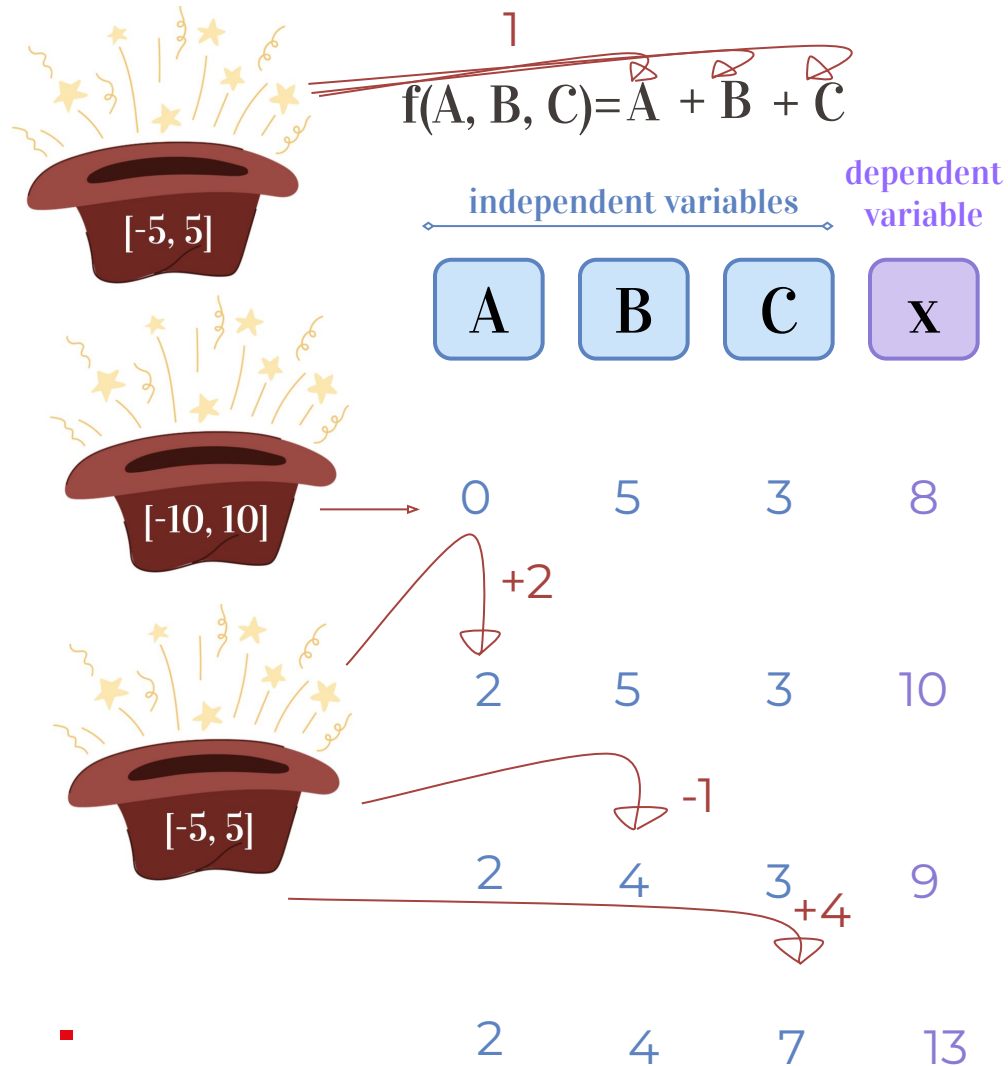
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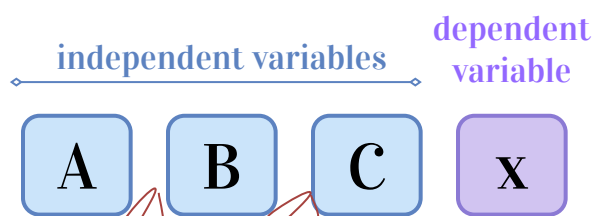




Linear Equations (linear_eqs)

27 equations
19 datapoints each
all linear
coefficients

**Methodology
: Dataset
Construction**



*

*

consider all
possible
permutations

$$A * B * C$$

Equations 1 (eqs_1)

27 equations
19 datapoints each
linear & nonlinear
no coefficients

Equations 2, 3 (eqs_2, eqs_3)

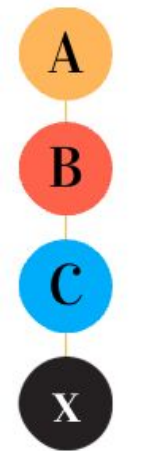
54 equations
19 datapoints each
linear & nonlinear
coefficients

Methodology : Dataset Construction



Methodology: Experiments Pipeline

RQ1: Baseline Performance

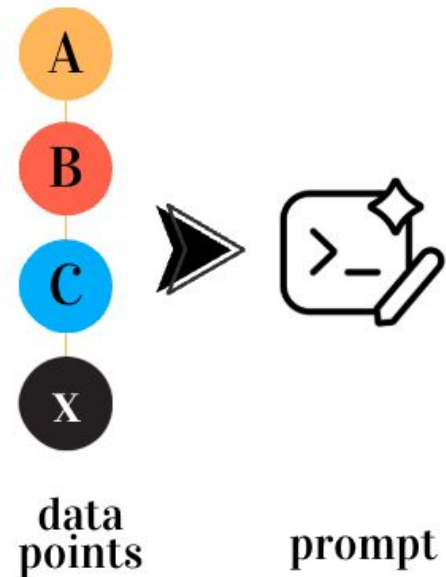


data
points

$D = \{(A_i, B_i, C_i, x_i)\}_{i=1}^n$ f : ground-truth symbolic equation s.t. $x = f(A, B, C)$

Methodology: Experiments Pipeline

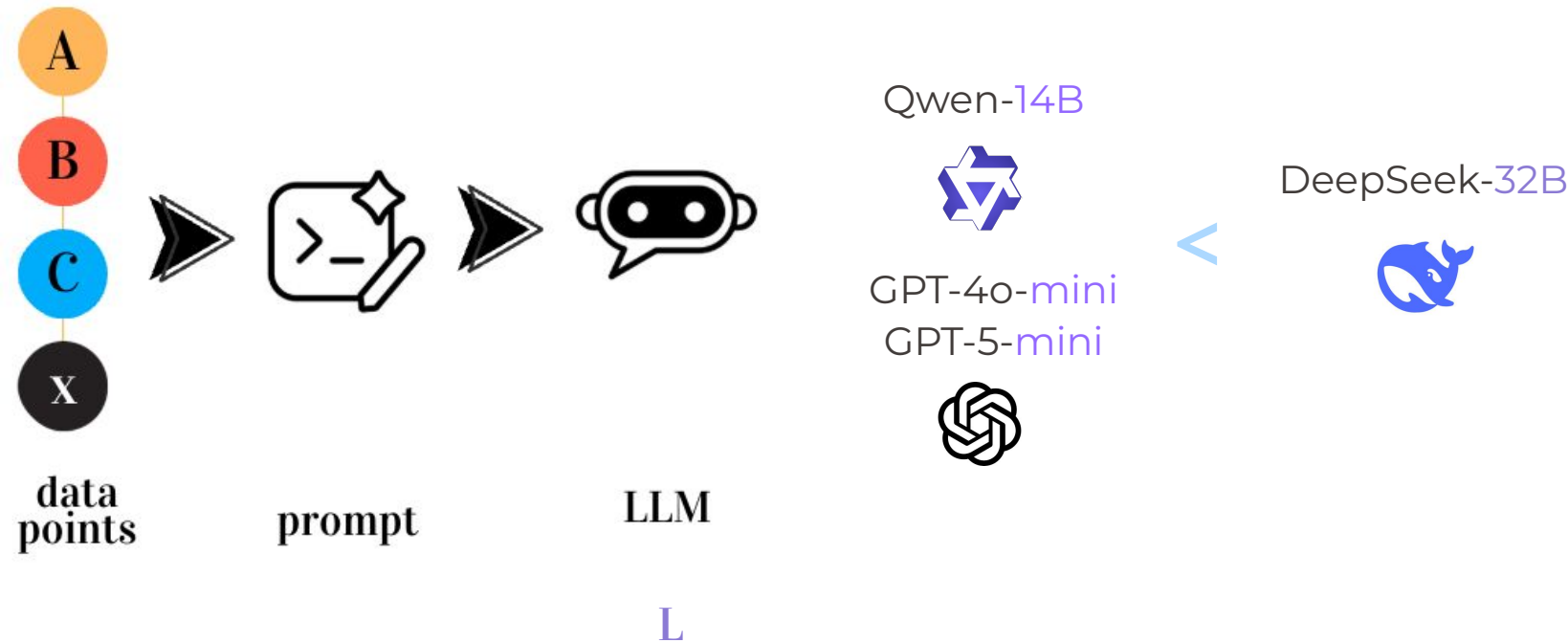
RQ1: Baseline Performance



P: task instruction, data points, ICL examples (optional)

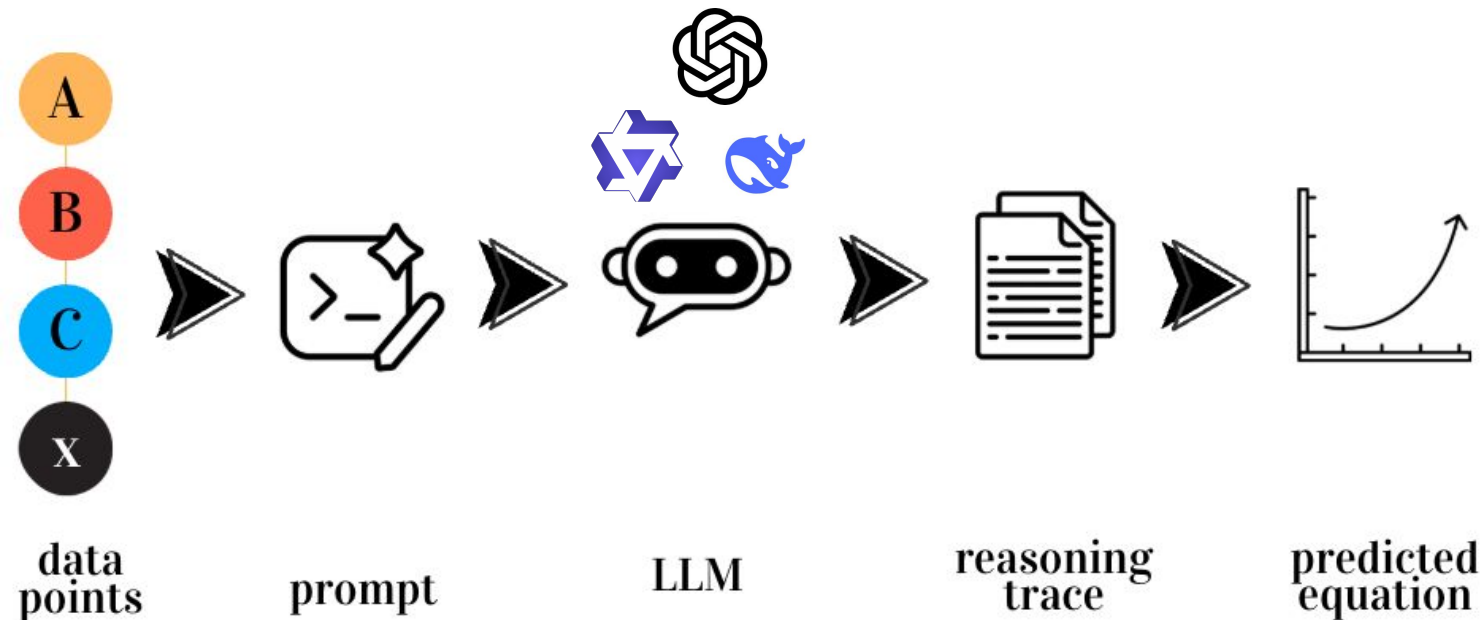
Methodology: Experiments Pipeline

RQ1: Baseline Performance



Methodology: Experiments Pipeline

RQ1: Baseline Performance

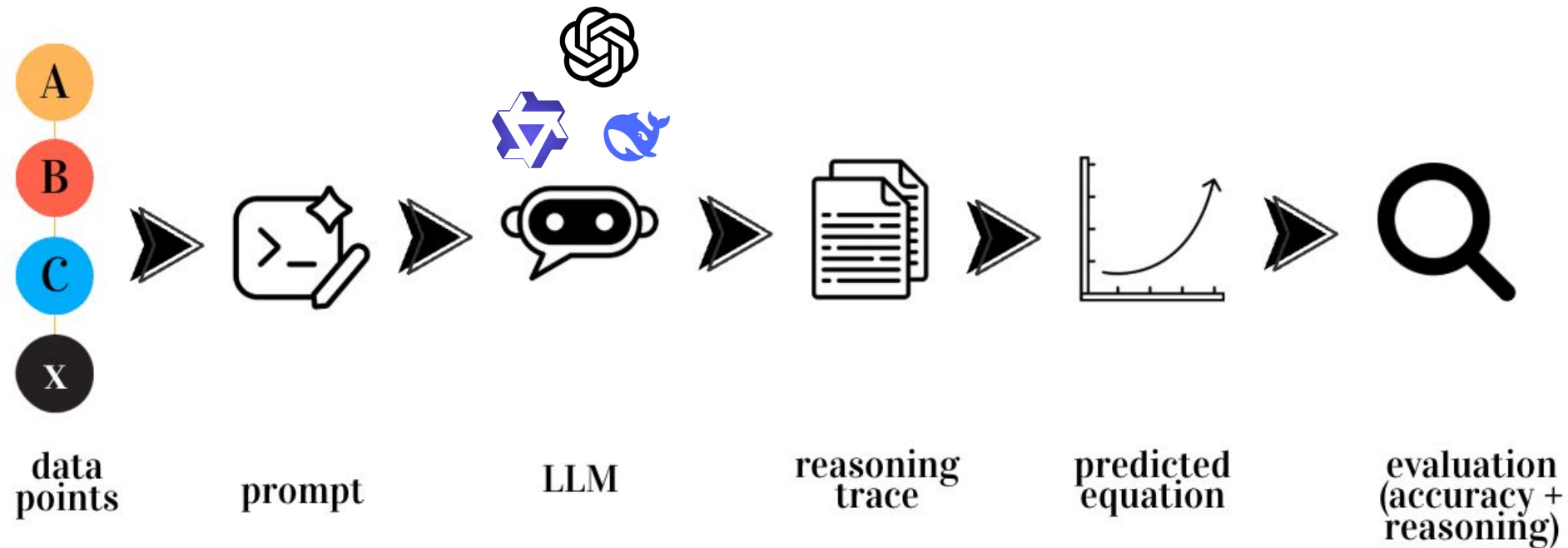


model output distribution: $\hat{y} \sim L(. | X, P)$

$\hat{y} = L(X, P)$: equation predicted by L

Methodology: Experiments Pipeline

RQ1: Baseline Performance



Symbolic accuracy: $\hat{f} \equiv f$

Does the predicted equation match the ground truth?

Methodology: Experimental Setup

RQ1: Baseline Performance

Prompting Styles

1. Direct
2. Chain-of-Thought (CoT) [10]

Conditions Tested

- Models with varying parameter size
- Equation complexity
- Varying number of data points

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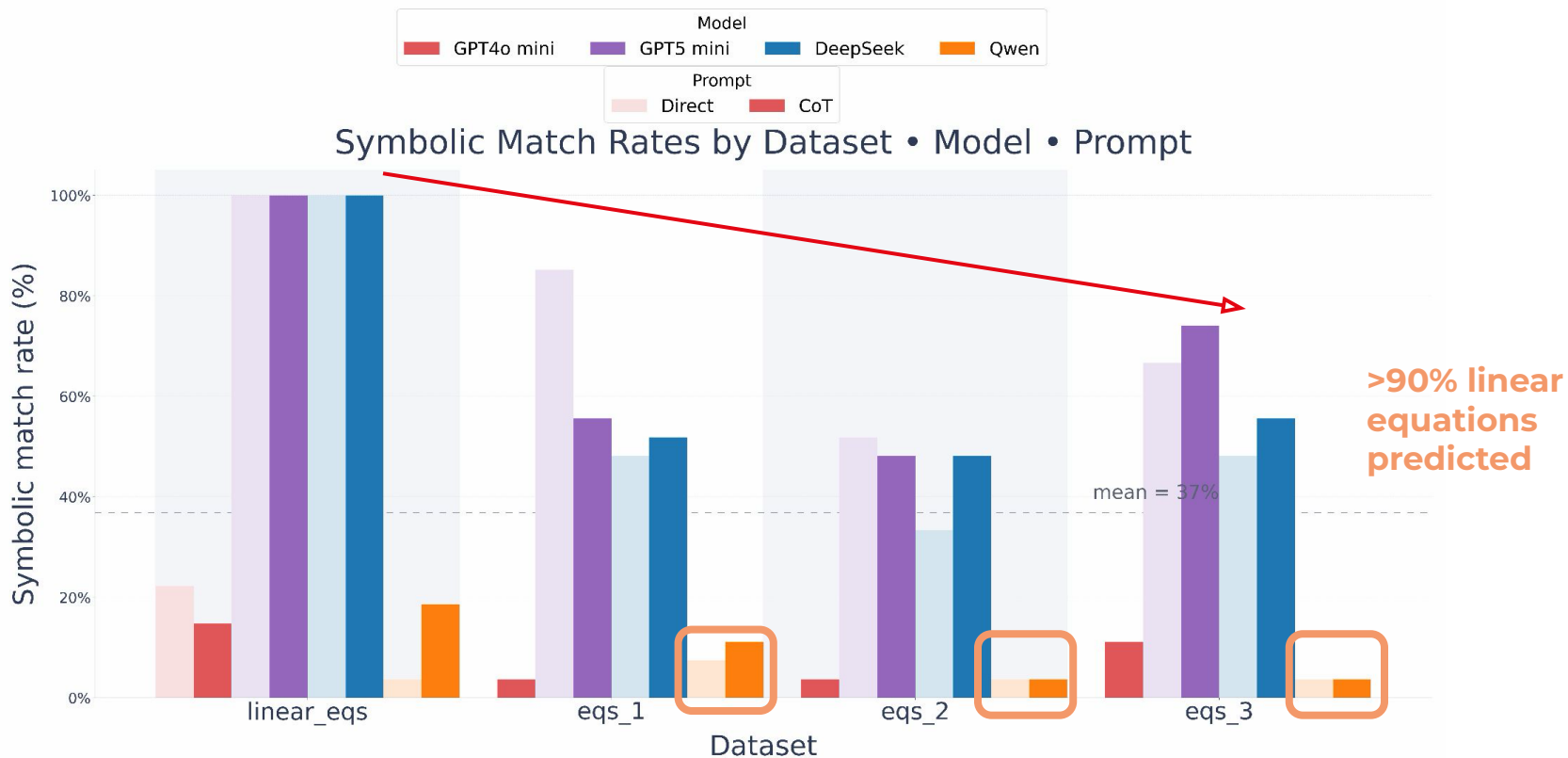


RQ2: In-Context Learning

How can **in-context learning (ICL)** and **structured reasoning strategies** be used to improve LLMs' inductive reasoning in numerical mathematical modeling?



RQ1: Baseline Performance



Results: Conclusions

RQ1: Baseline Performance

- **[TODO Add animation]**
- **Small LLMs (<32B)**
 - Fail at numerical modeling without guidance
- **Large LLMs**
 - Require >> output tokens
 - wrong initial hypotheses → cascading failure
- **Bias**
 - Default to **linear regression shortcuts**: >80% initial hypothesis is linear
- **Prompting: Chain-of-Thought (CoT)** yields best results
 - structured reasoning guidance
- **Limits**
 - Even large LLMs struggle with **non-linear/complex equations** and large coefficients.

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RQ2: In-Context Learning

How can **in-context learning (ICL)** and **structured reasoning strategies** be used to improve LLMs' inductive reasoning in numerical mathematical modeling?



Methodology: Stream of Search + Self-Backtracking

RQ2: In-Context Learning

- Encourages structured reasoning through in context examples
- Explicit reasoning structure exposes models to intermediate reasoning and revision steps
- Search over intermediate steps

```
- node: string index of the current node (e.g., "0", "0.0", "0.0.1", etc.)
  current_state:
    hypothesis: the current symbolic expression being tested (or null at the
                start)
    evidence: a list of verbal conclusions from earlier steps
    reasoning: a list of logical justifications for each step taken
  action: one of:
    - "<<EXPAND>>" (to add a new reasoning step)
    - "<<BACKTRACK>>" (to abandon the current path and return to a previous node)
    - "<<EXPAND_AFTER_BACKTRACK>>" (to try a new structure after backtracking)
    - "<<GOAL_CHECK>>" (to evaluate the final hypothesis on all datapoints)
  operation: what you're doing (e.g., "Compare dp1 and dp2 to isolate B")
  datapoints: the relevant datapoints being analyzed
  observation: what you observed from the changes (e.g. change in x with
               respect to a variable)
  updated_state: updated hypothesis, evidence, and reasoning after this step
    hypothesis: updated symbolic form for x (e.g., "x = A + 2*B + ___")
    evidence: cumulative evidence supporting this hypothesis
    reasoning: cumulative logical reasoning trace so far
  result: Whether the hypothesis fits all datapoints
  final_hypothesis: The final expression for x in terms of A, B, C, exactly
                   matching all datapoints
```

node per inductive step

backtrack / expand / expand after backtrack / goal check

Methodology: Designing ICL

Examples

ICL = in-context learning

1. **Zero shot** model responses
2. Identify **struggles**
 - a. assuming linear form
 - b. no goal check
3. **Hand Constructed** high quality examples
 - a. backtrack when revision of hypothesis necessary
 - b. test candidate equation on *all* data points

- **Full Task**
 - **Optimal** (linear) reasoning path: 5 equations
 - Nonlinear reasoning path, with **Backtracking**: 5 equations
 - Input data points included

Results: SoS + Backtracking

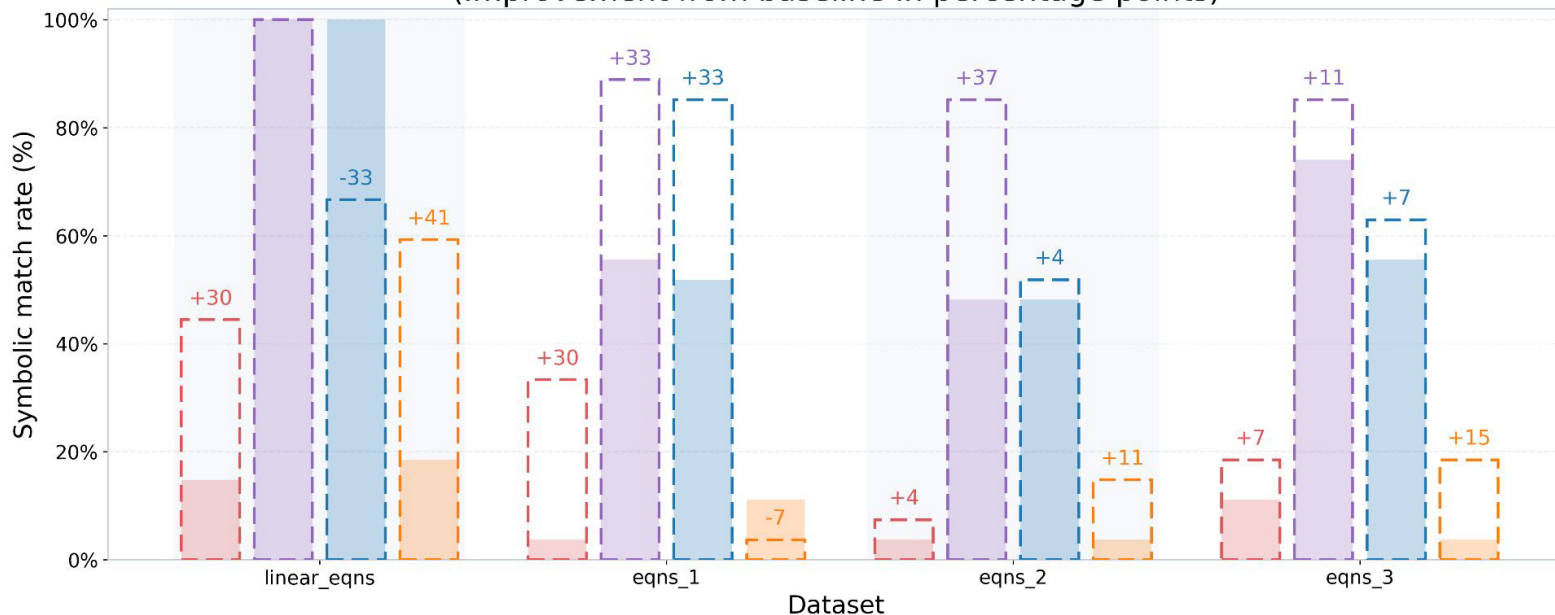
RQ2: In-Context Learning

improvement consistent across a majority of the models

Models + Method

GPT4o mini GPT5 mini DeepSeek Qwen CoT baseline ICL backtrack

Symbolic Match Rates by Dataset • Model • Method
(Improvement from baseline in percentage points)



Methodology: Subtask Decomposition

Goal: Identify bottlenecks for numerical mathematical modeling via inductive reasoning for LLMs

Subtask 1 (Cause-effect detection)

- How does each independent variable affect the independent variable x ?
- $+4 A \rightarrow +4 x$

Subtask 2 (Equation formation)

- Combine variable cause-effects into a full symbolic rule
- $+4 A \rightarrow +4 x$
- $+4 B \rightarrow +4 x$
- $+4 C \rightarrow +4 x$ linear: $x = A + B + C$

Subtask 1 + Subtask 2

- 5 equations of varying difficulty (linear and nonlinear)
- Inputs omitted due to context size limitations

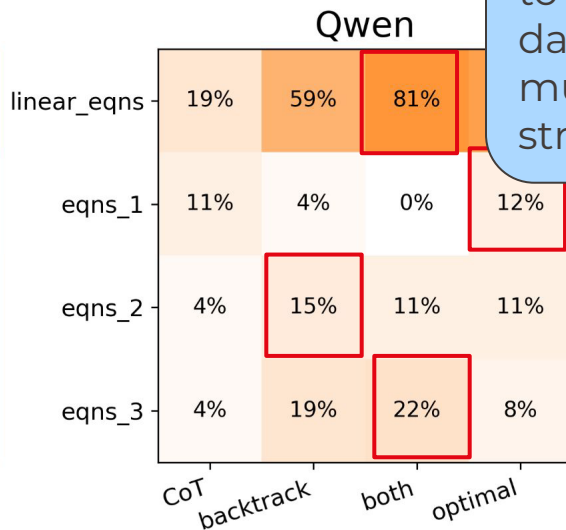
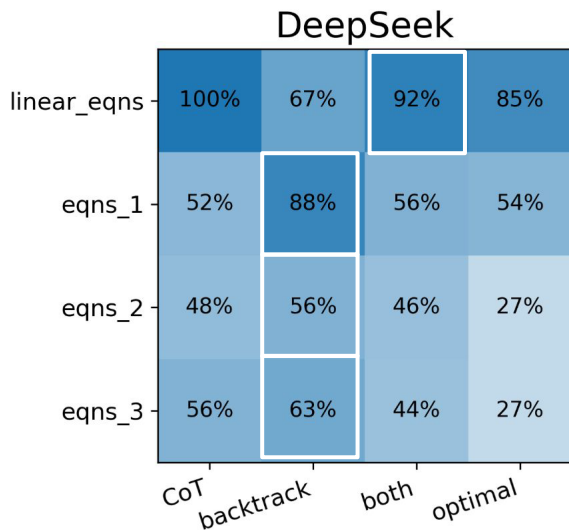
Results: Optimal vs Backtrack ICL Examples

RQ2: Baseline Performance

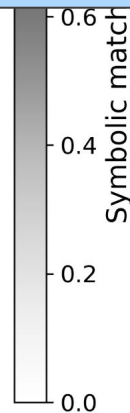
backtracking tends to do well for more complex equations

larger models make effective use of backtracking, smaller models get stuck

Effect of In-Context Learning Examples Reasoning Tra



“Linear model fails to fit multiple datapoints; explore multiplicative structure”



Research Questions

Goal: Investigate the **inductive reasoning** abilities of large language models (LLMs) in low-level numerical **mathematical modeling**.



RQ1: Baseline Performance

How well can LLMs reason inductively from numerical data **on their own**?



RQ3: Discoveries

Where do models **struggle** and what **reasoning bottlenecks** emerge?



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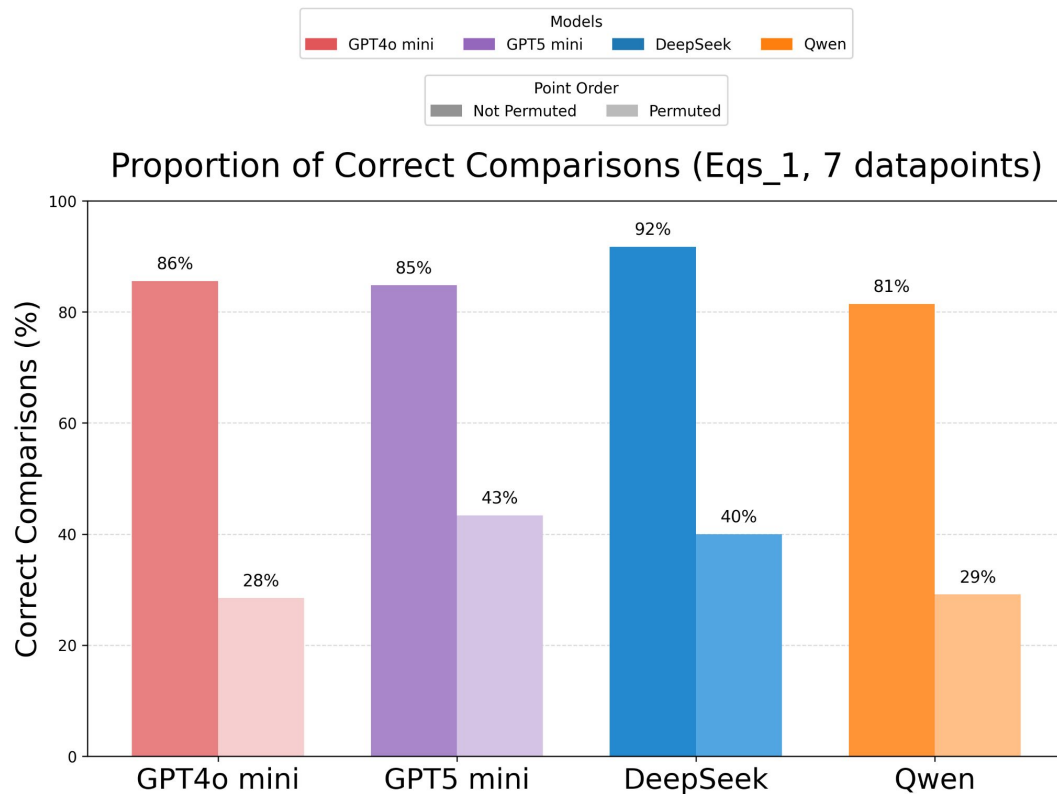


Results: Cause-Effect Detection

RQ2: In-Context Learning

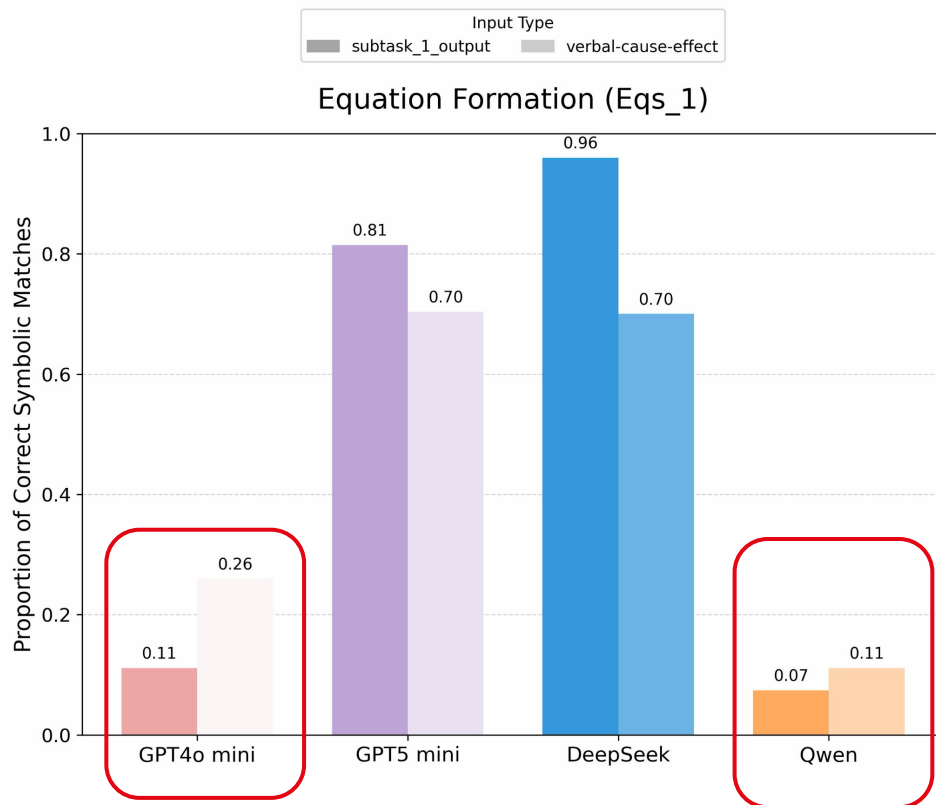
Models are
not robust to
varying input
order

When input is
structured,
models can
pattern
match
effectively



RQ2: In-Context Learning

Mathematical modeling
is a bottleneck for
smaller models,
especially for **nonlinear**
equation



Key Takeaways

RQ3

- **LLMs can perform interpretable, data-driven reasoning**
with backtracking + search strategies
 - Outperform direct & CoT prompts dramatically
- **Model size matters**
- **Biases and shortcuts:**
 - models default to linear regression
 - >90% Initial step: Let's assume $x = a*A + b*B + c*C + d$.
- **Equation complexity & order sensitivity**

- Fine-tune on high-quality reasoning traces.
- Probe internal mechanisms of good inductive reasoning LLMs
 - Discover underlying reasoning patterns
 - **Analyze Low-level:** attention patterns, pattern identification, arithmetic

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Caltech*



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Sreeyutha Ratala